

On the Connectivity of a Graph and its Complement Graph

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Abstract

In this paper, we study the connectivity of the complement of a graph. We first establish a simple formula to compute the connectivity of a graph that comes from the join of graphs. Based on this formula, we give an upper and a lower bound to the connectivity of the complement of a graph.

Keyword: connectivity, complement, join, vertex of attachment.

1 Introduction

All graphs considered in this paper are finite and simple (i.e., without loops and multiple edges). We denote the vertex set and edge set of a graph G by $V(G)$ and $E(G)$, respectively. The *order* of G is defined to be $n(G) = |V(G)|$. The *complement* of G is the graph denoted by $\overline{G}$ such that $V(\overline{G}) = V(G)$ and $E(\overline{G}) = \{uv : uv \notin E(G)\}$. For a subset $S \subset V(G)$, the subgraph of G induced by S is denoted by $G[S]$. Also, we use $G - S$ to denote the graph $G[V \setminus S]$ where $V \setminus S$ is the set $\{v \in V \mid v \notin S\}$. Similarly, for $F \subset E(G)$ we denote the graph with vertex set $V(G)$ and edge set $E(G) \setminus F$ by $G - F$. If S or F is a singleton x , we use the notation $G - x$ instead of $G - \{x\}$. If H is a subgraph of G , then for simplicity, we sometimes write H to mean $V(H)$.

A graph G is *connected* if there is at least one path between any two vertices of G , otherwise G is *disconnected*. The *vertex-connectivity* (or *connectivity* for simplicity) of a graph G , denoted by

$\kappa(G)$, is the minimum number of vertices whose removal from G results in a disconnected or trivial graph, i.e., a graph has only one vertex. Thus, $\kappa(K_n) = n - 1$, where K_n is a complete graph with n vertices. A graph G is *k-connected* if $\kappa(G) = k$, or equivalently, it has at least $k + 1$ vertices and for every pair of nonadjacent vertices u and v there are at least k internally-vertex-disjoint paths joining u and v in G .

In [1], Ando defined that a graph G is *critical connected* if $\kappa(G - v) = \kappa(G) - 1$ for each vertex $v \in V(G)$ and *(k, k')-critical connected* if $\kappa(G - v) = k - 1$ or $\kappa(\overline{G} - v) = k' - 1$ for each vertex $v \in V(G)$, where $k = \kappa(G)$ and $k' = \kappa(\overline{G})$. As a result, Ando obtained an upper bound of the number of vertices for *(k, k')-critical connected* graphs using the terms k and k' . Ando and Kaneko [2] defined that a graph G is *bi-3-connected* if both G and $\overline{G}$ are 3-connected. As a consequence, they proved that every bi-3-connected graph of order at least 22 has a two-vertex set called *bi-contractible pair* whose contraction results still in a bi-3-connected graph. Recently, Ito and Yokoyama [4] showed that $\kappa(\overline{G})$ can be computed in linear time if $\kappa(\overline{G}) \leq 3$. In this paper, we shall propose an upper bound and a lower bound of $\kappa(\overline{G})$ related to $\kappa(G)$.

The rest of this paper is organized as follows. In Section 2, we give some definitions and notation. Furthermore, some prior results are introduced. In Section 3, we discuss the connectivity of a graph which is obtained from the join of graphs. In Section 4, we give an upper and a lower bound to the connectivity of the complement of a graph. Finally, some concluding remarks are given in Section 5.

2 Preliminaries

A subgraph H of a graph G is called a *component* if H is a maximal connected subgraph of G . A *cutset* of G is a subset of vertices whose removal either increases the number of components or increases the number of isolated vertices in G . A cutset with the minimum size is called a *minimum cutset*. Using this modification of the standard definition permits one to compute $\kappa(G)$ by finding the size of a minimum cutset for any connected graph G . Specifically, a cutset with only one vertex is called a *cut vertex*.

Let G_1 and G_2 be two vertex-disjoint graphs. Then the *join*, $G_1 \vee G_2$, of G_1 and G_2 is the graph with vertex set $V(G_1) \cup V(G_2)$ and edge set $E(G_1) \cup E(G_2) \cup \{uv : u \in V(G_1) \text{ and } v \in V(G_2)\}$. Obviously, it follows from the definition that $G_1 \vee G_2$ is connected. For convenience, we write $\bigvee_{i=1}^k G_i = G_1 \vee G_2 \vee \dots \vee G_k$. We use an example to depict the concept of the join of two graphs. Figure 1(a) shows that two separated graphs G_1 and G_2 , and Figure 1(b) shows the graph $G_1 \vee G_2$. Note that both G_1 and G_2 are subgraphs of $G_1 \vee G_2$ induced by $V(G_1)$ and $V(G_2)$, respectively.

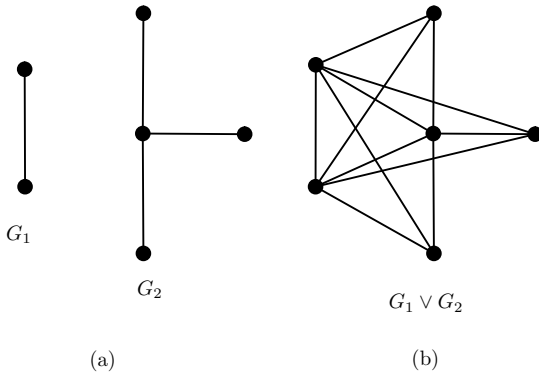


Figure 1: Two graphs and their join.

For a subgraph H of G , the *attachment* of H in G , denoted by $A_G(H)$, is the maximal subset of $V(H)$ such that every vertex in the set is adjacent to at least one vertex of $G - H$. Thus, $A_G(G) = A_G(\emptyset) = 0$. For two nonempty subsets $S_1, S_2 \subset V(G)$, the set containing all edges of G with one end vertex in S_1 and the other end vertex in S_2 is denoted by $E_G[S_1, S_2]$. When no ambiguity arises, we drop the index G in the above notations.

Boland et al. in [3] studied the *inclusive connectivity* (*i-connectivity* for short) of graphs, which is

a variant pertaining to the connectivity of graphs. Let G be a graph and $v \in V(G)$. The *vertex i-connectivity* of v , denoted by $\kappa_i(v, G)$, is the cardinality of a minimum vertex set $T \subset V(G)$ whose removal results in a subgraph containing v as a cut vertex. In particular, $\kappa_i(v, G) = 0$ if v is a cut vertex of G , and $\kappa_i(v, G) = n(G) - 1$ if T does not exist. As a result, Boland et al. provided a way to find the vertex i -connectivity of the join of two graphs.

Theorem 1 [3] *Let G and H be two arbitrary connected graphs and $v \in V(G)$. Then*

$$\kappa_i(v, G \vee H) = \min\{\kappa_i(v, G) + n(H), \kappa(H) + n(G) - 1\}.$$

For example, Figure 2 shows a graph $C_4 \vee K_2$, where C_4 contains a vertex v . From Theorem 1, we are easy to verify that $\kappa_i(v, C_4) = 1$, $n(K_2) = 2$, $\kappa(K_2) = 1$, and $n(C_4) = 4$. Thus $\kappa_i(v, C_4 \vee K_2) = 3$. In this figure, the part of shadowed vertices is the minimum vertex set T such that v becomes a cut vertex if T is removed.

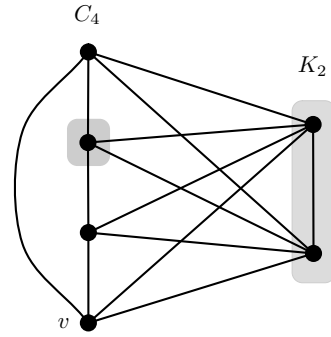


Figure 2: $\kappa_i(v, C_4 \vee K_2) = 3$ where $v \in V(G)$.

3 The connectivity of the join of graphs

In this section, we discuss the connectivity of the join of graphs. The following lemma is similar to Theorem 1 that gives a formula to compute the connectivity of the join of two graphs.

Lemma 2 *Let G_1 and G_2 be two arbitrary graphs that each has a minimum cut set C_i for $i = 1, 2$, respectively. Then*

$$\kappa(G_1 \vee G_2) = \min\{n(G_1) + \kappa(G_2), n(G_2) + \kappa(G_1)\}.$$

In particular, if $\kappa(G_1 \vee G_2) = n(G_i) + \kappa(G_j)$ where $i, j \in \{1, 2\}$ and $i \neq j$, then $V(G_i) \cup C_j$ is a minimum cut set of $G_1 \vee G_2$.

Proof. If both G_1 and G_2 are complete graphs, then so is $G_1 \vee G_2$. In this case, it is clear that $\kappa(G_1 \vee G_2) = n(G_1) + n(G_2) - 1$ and the lemma holds. We now suppose that at least one of G_1 and G_2 is not a complete graph and let C be a minimum cut set in $G_1 \vee G_2$. Then $\kappa(G_1 \vee G_2) = |C|$. Since in $G_1 \vee G_2$ every vertex of G_1 is adjacent to every vertex of G_2 , if two vertices u and v are separated in $(G_1 \vee G_2) - C$, then both u and v must come from G_1 or G_2 . Thus, either $V((G_1 \vee G_2) - C) \subset V(G_1)$ or $V((G_1 \vee G_2) - C) \subset V(G_2)$, which is dependent on the size of C .

Without loss of generality, we assume $V((G_1 \vee G_2) - C) \subset V(G_1)$. Clearly, $V(G_2) \subset C$. Since $G_1 \vee G_2$ is not a complete graph, $(G_1 \vee G_2) - C$ has at least two components that each contains only the vertices from G_1 . This implies that C must contain a cut set of G_1 . Thus, $|C| \geq n(G_2) + \kappa(G_1)$. Since C is a minimum cut set in $G_1 \vee G_2$, it must consist of exactly $V(G_2)$ and a minimum cut set of G_1 . In particular, we can choose $C = V(G_2) \cup C_1$. $\square$

We now use an example to illustrate Lemma 2. Two graphs G_1 and G_2 are shown in Figure 2(a), and $G_1 \vee G_2$ is shown in Figure 2(b). Since $n(G_1) = 3$, $\kappa(G_1) = 1$, $n(G_2) = 5$, and $\kappa(G_2) = 2$, by Lemma 2, $\kappa(G_1 \vee G_2) = \min\{5 + 3, 3 + 2\} = 5$. The shadowed part in each graph indicates the minimum cutset of G_1 , G_2 , and $G_1 \vee G_2$, respectively.

According to Lemma 2, the following extension can easily be proved by induction.

Theorem 3 Let $G_1, G_2, \dots, G_n$ be distinct graphs that each has a minimum cut set C_i for $i = 1, 2, \dots, n$. Then

$$\kappa\left(\bigvee_{i=1}^n G_i\right) = \min_{1 \leq j \leq n} \left\{ \kappa(G_j) + \sum_{\substack{1 \leq i \leq n \\ i \neq j}} n(G_i) \right\}$$

In particular, if $\kappa(\bigvee_{i=1}^n G_i) = \kappa(G_j) + \sum_{\substack{1 \leq i \leq n \\ i \neq j}} n(G_i)$ for some $j \in \{1, \dots, n\}$, then $(\bigcup_{\substack{1 \leq i \leq n \\ i \neq j}} V(G_i)) \cup C_j$ is a minimum cut set of $\bigvee_{i=1}^n G_i$.

Proof. We prove this theorem by induction on n . For $n = 2$, the result is shown in Lemma 2. Assume that this theorem holds for any k graphs $G_1, G_2, \dots, G_k$, where $k \geq 2$. We now consider

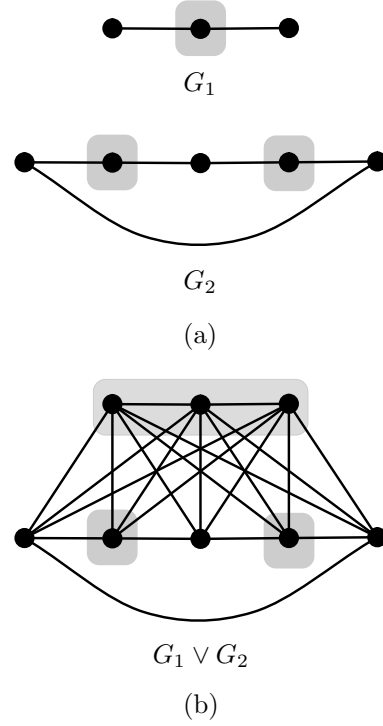


Figure 3: The minimum cutsets of G_1 , G_2 and $G_1 \vee G_2$.

$k + 1$ graphs $G_1, G_2, \dots, G_{k+1}$ as follows.

$$\begin{aligned} & \kappa\left(\bigvee_{i=1}^{k+1} G_i\right) \\ &= \kappa\left(\left(\bigvee_{i=1}^k G_i\right) \vee G_{k+1}\right) \\ &= \min\left\{n\left(\bigvee_{i=1}^k G_i\right) + \kappa(G_{k+1}), n(G_{k+1}) + \kappa\left(\bigvee_{i=1}^k G_i\right)\right\} \\ &= \min\left\{\sum_{i=1}^k n(G_i) + \kappa(G_{k+1}), n(G_{k+1}) + \min_{1 \leq j \leq k} \left\{ \kappa(G_j) + \sum_{\substack{1 \leq i \leq k \\ i \neq j}} n(G_i) \right\}\right\} \\ &= \min\left\{\sum_{i=1}^{k+1} n(G_i) + \kappa(G_{k+1}) - n(G_{k+1}), \sum_{i=1}^{k+1} n(G_i) + \min_{1 \leq j \leq k} \left\{ \kappa(G_j) - n(G_j) \right\}\right\} \\ &= \sum_{i=1}^{k+1} n(G_i) + \min\left\{\kappa(G_{k+1}) - n(G_{k+1}), \min_{1 \leq j \leq k} \left\{ \kappa(G_j) - n(G_j) \right\}\right\} \end{aligned}$$

$$\begin{aligned}
&= \sum_{i=1}^{k+1} n(G_i) + \min_{1 \leq j \leq k+1} \{\kappa(G_j) - n(G_j)\} \\
&= \min_{1 \leq j \leq k+1} \{\kappa(G_j) + \sum_{\substack{1 \leq i \leq k+1 \\ i \neq j}} n(G_i)\}
\end{aligned}$$

This shows that the connectivity of $\bigvee_{i=1}^n G_i$ is $\kappa(G_j) + \sum_{\substack{1 \leq i \leq n \\ i \neq j}} n(G_i)$ for some $j \in \{1, \dots, n\}$.

In this case, to find out a minimum cutset C of $\bigvee_{i=1}^{k+1} G_n$, we can choose $C = (\bigcup_{\substack{1 \leq i \leq n \\ i \neq j}} V(G_i)) \cup C_j$.

This completes the proof. $\square$

Corollary 4 *If G is a disconnected graph with k components, then $\kappa(\overline{G}) \geq k - 1$.*

Proof. Let $H_1, H_2, \dots, H_k$ be the k components of G . Clearly, $\overline{G} = \bigvee_{i=1}^k \overline{H_i}$ is connected. By Theorem 3 and the fact that $\kappa(\overline{H_i}) \geq 0$ and $n(\overline{H_i}) \geq 1$ for every $i = 1, \dots, k$, we have $\kappa(\overline{G}) = \kappa(\bigvee_{i=1}^k \overline{H_i}) = \min_{1 \leq j \leq k} \{\kappa(\overline{H_j}) + \sum_{\substack{1 \leq i \leq k \\ i \neq j}} n(\overline{H_i})\} \geq k - 1$. $\square$

4 The connectivity of the complement of a Graph

Throughout the rest, we consider that G is a nontrivial, noncomplete, connected graph with a cutset C , and suppose that there are $k \geq 2$ components in $G - C$, say H_i for $i = 1, \dots, k$. Clearly, $\overline{G - C} = \overline{H_1} \vee \overline{H_2} \vee \dots \vee \overline{H_k}$ and the attachment $A_G(H_i)$, $i = 1, \dots, k$, are nonempty sets. For simplicity, we write A_i to mean $A_G(H_i)$ and let $D_i = V(H_i - A_i)$.

With respect to the cutset C of G , we first define a special graph $\overline{G_C}$. Then, we will show that $\kappa(\overline{G_C})$ is a lower bound of the connectivity of $\overline{G}$ and it can be computed by a simple way. Formally, we define as follows:

$$\overline{G_C} = \overline{G} - E_{\overline{G}}[C, \bigcup_{i=1}^k A_i] - (\bigcup_{i=1}^k E_{\overline{G}}[A_i, D_i]) \quad (1)$$

Note that $\overline{G_C}$ is a spanning subgraph of $\overline{G}$. For example, a visual illustration of a graph G with a cutset C is shown in Figure 4(a), where $G - C$ contains three components H_1, H_2, H_3 and the edge sets $E[C, \bigcup_{i=1}^3 A_i]$ and $E[A_i, D_i]$ for $1 \leq i \leq 3$ are drawn by dotted lines. According to the definition (1), another visual illustration of the graph $\overline{G_C}$ is shown in Figure 4(b), where the edge sets $E_{\overline{G_C}}[\overline{H_i}, \overline{H_j}] = \{xy : x \in V(H_i), y \in V(H_j)\}$ for

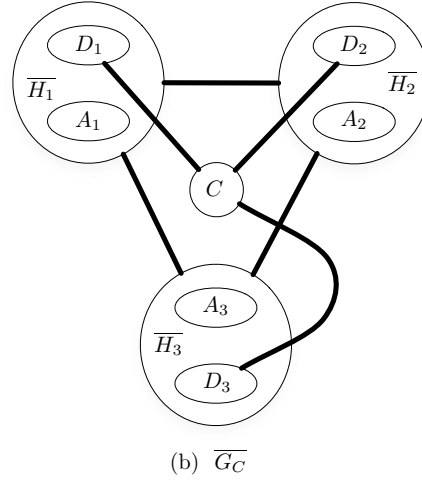
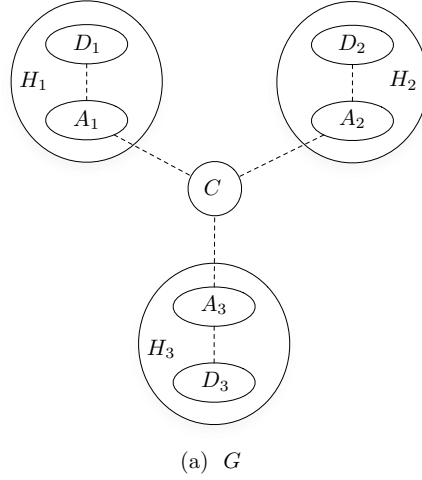


Figure 4: (a) A visualization of a graph G with a cutset C and the three components of $G - C$; (b) The graph $\overline{G_C}$.

$i \neq j$ and $E_{\overline{G_C}}[C, D_i] = \{cd : c \in C, d \in D_i\}$ for $1 \leq i \leq 3$ are drawn by solid lines.

Lemmas 5 and 6 are used to achieve $\kappa(\overline{G_C})$.

Lemma 5 *Let G be a graph and H, K be two disjoint connected induced subgraphs of G . Suppose that K has a minimum cutset C_K . Then, the following statements are true:*

- (1) *If $\kappa(G[K \cup H]) > \kappa(K)$, then $E[S, H] \neq \emptyset$, where S is any component of $K - C_K$.*
- (2) *If $\kappa(G[K \cup H]) \leq \kappa(K)$, then a minimum cutset of $G[K \cup H]$ is either C_K or a subset of $A_{G[K \cup H]}(K) \cup V(H)$.*

Proof. To prove statement (1), we let S be any component of $K - C_K$. Since $\kappa(G[K \cup H]) > \kappa(K)$, $G[K \cup H] - C_K$ is connected and it contains S as an induced subgraph. In this case, since there exist no edges joining vertices between any two components of $K - C_K$, S must be directly connected to H . Thus $E[S, H] \neq \emptyset$.

We prove statement (2) by contradiction. Suppose to the contrary that $\kappa(G[K \cup H]) \leq \kappa(K)$ and there is a minimum cutset C of $G[K \cup H]$ which is not C_K or a subset of $A_{G[K \cup H]}(K) \cup V(H)$. Let

$$X = C - (A_{G[K \cup H]}(K) \cup V(H))$$

and

$$Y = C - X.$$

Clearly, $X \neq \emptyset$ and Y is not a cutset of $G[K \cup H]$. Thus $G[K \cup H] - Y$ is connected. Let

$$Z = A_{G[K \cup H]}(K) - Y.$$

Since $A_{G[K \cup H]}(K)$ is a cutset of $G[K \cup H]$, Y cannot contain all the vertices of $A_{G[K \cup H]}(K)$. This implies that Z is nonempty. By the fact $C = X \cup Y$, X is a cutset of $G[K \cup H] - Y$. Moreover, we can find that the vertices of $(H - Y) \cup Z$ must be contained in one of the components of $G[K \cup H] - Y - X$. Therefore, X is also a cutset of $K - Y$. Now, if $K[C]$ is a cutset of K and $K[C] \neq C_K$, then $n(K[C]) > \kappa(K)$. However, by assumption $\kappa(K) \geq \kappa(G[K \cup H]) = n(C) \geq n(K[C])$. It is a contradiction. $\square$

Lemma 6 For any graph G with a cutset C ,

$$\kappa(\overline{G_C}) = \min\left\{\sum_{i=1}^k n(D_i), \min_{1 \leq i \leq k} \{n(G - C - H_i)\}\right\}$$

Proof. Let $H = \overline{G - C} = \bigvee_{i=1}^k \overline{H_i}$ and C_H be a minimum cutset of H . By Theorem 3, all components of $H - C_H$ must be contained in $\overline{H_j}$ for a certain $j \in \{1, \dots, k\}$. In particular, there is a component of $H - C_H$, say S , that contains only the vertices of A_j since by definition $\kappa(\overline{H_j}) = 0$ and $E_{\overline{G_C}}[A_j, D_j] = \emptyset$.

Obviously, $\overline{G_C} = \overline{G_C}[V(H) \cup C]$. We now claim that $\kappa(\overline{G_C}) \leq \kappa(H)$. Suppose not, by Lemma 5 we know that every component of $H - C_H$ is directly connected to C . Thus $E_{\overline{G_C}}[S, C] \neq \emptyset$. However, this is impossible since S contains only the vertices of A_j and by definition there exist no edges between A_j and C in $\overline{G_C}$. Since $\kappa(\overline{G_C}) \leq \kappa(H)$,

again by Lemma 5, a minimum cutset of $\overline{G_C}$ is either C_H or a subset of $A_{\overline{G_C}}(H) \cup C$. Thus we consider the following two cases.

Case 1: C_H is a minimum cutset of $\overline{G_C}$. In this case $\kappa(\overline{G_C}) = \kappa(H) = |C_H|$. By Theorem 3, we can compute the connectivity of $\overline{G_C}$ as follows.

$$\begin{aligned} \kappa(H) &= \kappa(\bigvee_{i=1}^k \overline{H_i}) \\ &= \min_{1 \leq j \leq k} \{\kappa(\overline{H_j}) + \sum_{\substack{1 \leq i \leq k \\ i \neq j}} n(\overline{H_i})\} \\ &= \min_{1 \leq i \leq k} \{n(G - C - H_i)\} \end{aligned} \quad (2)$$

The last equality holds due to the fact that $\kappa(\overline{H_j}) = 0$ and $n(H) - n(\overline{H_i}) = n(G - C - H_i)$.

Case 2: There is a subset of $A_{\overline{G_C}}(H) \cup C$, say T , which forms a minimum cutset of $\overline{G_C}$. Since each vertex of C is adjacent to each vertex of D_i in $\overline{G_C}$ for all $i = 1, \dots, k$, either $C \subseteq T$ or $(\bigcup_{i=1}^k D_i) \subseteq T$. For the former case, since $T - C$ is a cutset of H , it follows from Case 1 that $|T| \geq |C| + \kappa(H) = |C \cup C_H| > \kappa(\overline{G_C})$, a contradiction. Contrastively, since $E_{\overline{G_C}}[\bigcup_{i=1}^k A_i, C] = \emptyset$ and T is a minimum cutset of $\overline{G_C}$, the latter case implies that $T = \bigcup_{i=1}^k D_i$ and $\kappa(\overline{G_C}) = |T| = \sum_{i=1}^k n(D_i)$. $\square$

As a consequence, we have the following result.

Theorem 7 Let G be an arbitrary graph with a minimum cutset C and $\overline{G_C}$ be defined as in (1). Then

$$\kappa(\overline{G_C}) \leq \kappa(\overline{G}) \leq \kappa(\overline{G - C}) + \kappa(G)$$

where,

$$\kappa(\overline{G - C}) = \min_{1 \leq i \leq k} \{n(G - C - H_i)\}.$$

Proof. The lower bound of $\kappa(\overline{G})$ follows from Lemma 6 and the fact that $\overline{G_C}$ is a spanning subgraph of $\overline{G}$. Similarly, since $\overline{G}$ is a spanning subgraph of $\overline{G - C} \vee \overline{C}$, $\kappa(\overline{G}) \leq \kappa(\overline{G - C} \vee \overline{C})$. By Theorem 3,

$$\begin{aligned} \kappa(\overline{G}) &\leq \min\{\kappa(\overline{G - C}) + n(\overline{C}), \kappa(\overline{C}) + n(\overline{G - C})\} \\ &\leq \kappa(\overline{G - C}) + n(C) \\ &= \kappa(\overline{G - C}) + \kappa(G). \end{aligned}$$

Moreover, $\kappa(\overline{G - C})$ can be computed as in (2). $\square$

5 Concluding remarks

In this paper, we derive an upper bound and a lower bound of the connectivity of the complement of a graph G , i.e., $\kappa(\overline{G-C}) \leq \kappa(\overline{G}) \leq \kappa(G-C) + \kappa(G)$, where $\kappa(\overline{G-C})$ and $\kappa(G-C)$ can be computed by a simple way. We are now trying to find the class of graphs so that the equality holds in the above inequality.

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